Andreas Tzitzikas

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Education

University of Maryland, College Park, MD — Expected May 2027

Master of Science in Computer Engineering — **GPA: 4.00**

Bachelor of Science in Computer Engineering (GPA: 3.88) — Expected May 2026

Relevant Coursework: Digital CMOS VLSI Design (ENEE640), CMOS VLSI Design (ENEE646), Introduction to Micro/Nanoelectronics (ENEE304), Elements of Discrete Signal Analysis (ENEE222), Computer Organization (ENEE350)

Projects

6-Bit Dadda Multiplier | CMOS VLSI Design, Neural Network Accelerators — Spring 2025

- Designed 6-bit Dadda multiplier for neural network accelerators, achieving 307 ps propagation delay and 0.279 mW power consumption through CMOS optimization.
- Implemented Dadda architecture with partial product generation, reduction stages, and RCA final summation, including cheating device for worst-case all-ones inputs.
- Optimized transistor sizing (NMOS M=15, PMOS M=14, 300nm gate width) and minimized parasitic capacitance for maximum speed-power efficiency.

Tomasulo Out-of-Order RISC-V CPU | CMOS VLSI Design, ASIC Implementation — Nov 2025 - Dec 2025

- Architected CMOS VLSI system using Tomasulo's Algorithm with register renaming, reservation stations, and reorder buffer for advanced semiconductor design.
- Achieved 36% higher operating frequency (69.7MHz vs 51.2MHz) through CMOS circuit optimization, timing closure, and power-performance trade-offs.
- Synthesized to fabrication-ready GDSII using Sky130 130nm CMOS PDK with zero DRC/LVS violations and comprehensive static timing analysis.

5-Stage In-Order RISC-V CPU | CMOS Circuit Design, ASIC Synthesis — Aug 2025 - Nov 2025

- Designed synchronous CMOS digital circuits implementing hazard detection, data forwarding, and branch resolution for reliable semiconductor operation.
- Optimized critical path timing through CMOS transistor sizing, clock domain management, and synchronous reset implementation for manufacturable ASIC designs.
- Synthesized CMOS standard cells achieving 10,121 total cells with 1.0mm^2 die area, demonstrating PPA optimization in 130nm CMOS technology.
- Implemented comprehensive verification methodology ensuring 100% functional coverage and timing closure for CMOS circuit reliability.

Experience

Embedded Systems Intern, Alchemity – College Park, MD — Jun 2025 – Aug 2025

- Accelerated validation of a hybrid solid oxide fuel cell reactor by developing a full-stack automation suite (**Rust**, **Python**) to parallelize testing, converting multi-day manual experiments into overnight runs to significantly increase throughput.
- Engineered real-time **STM32** firmware to precisely control material synthesis within modular reactors, featuring non-blocking motor/relay drivers and **SPI** peripheral management.
- Built end-to-end control interfaces including desktop GUIs, embedded display drivers, and CLI tools to streamline workflows between hardware and software teams.

Technical Coordinator, Electronics Prototyping Lab, Terrapin Works – College Park, MD — Jan 2024 – Present

- Provide end-to-end **PCB** support (schematic to assembly) for 20+ students and faculty per semester, maintaining **LPKF** tools to ensure **90% uptime**.
- Lead training for new employees and expand campus outreach to increase awareness of PCB services, while assisting researchers with diagnostics.

Skills

CMOS VLSI Design: Digital CMOS Design, Transistor-Level Design, Standard Cell Libraries, CMOS PDKs
Circuit Design: Synchronous Digital Circuits, CMOS Logic Design, Timing Analysis, Power Analysis
Semiconductor Tools: Cadence Spectre, CMOS PDKs (Sky130), DRC/LVS Verification, Static Timing Analysis
ASIC Methodologies: PPA Optimization, CMOS Synthesis, Physical Design, Semiconductor Manufacturing